

EVA

Machine Learning Reasoning Layer

Master Product Requirements Document

Version 1.0 | Internal Engineering Reference

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This document is the master reference for EVA's Machine Learning Reasoning Layer (MLRL). It defines the system's design philosophy, pipeline architecture, Global Analysis Ledger structure, cross-module rules, and links to each module's individual specification.

Module documents referenced by this master:

- PFTDC — Problem Framing & Training Data Construction
- MCG — Model Candidate Generator
- TEM — Training & Evaluation Module
- RVEM — Reliability & Validation (Explainability) Module
- PMDD — Prediction, Monitoring & Dockerized Deployment

1. Purpose & Scope

The Machine Learning Reasoning Layer (MLRL) is EVA's capability for learning predictive patterns from data when analytical reasoning alone cannot answer the user's objective. It is a supporting layer — not the primary intelligence.

EVA's reasoning order is fixed:

| *Understanding → Reasoning → Hypothesis → Evidence → Then Learning*

The MLRL activates only when the Global Analysis Ledger determines that a future outcome must be predicted, a classification decision is required, or pattern generalization cannot be achieved through descriptive analysis alone. If these conditions are not met, EVA completes the analysis without building a model.

2. Design Philosophy

Traditional ML systems begin with models and search for meaning later. EVA reverses this. Machine learning in EVA is contextual, explainable, and evidence-bound. The model is treated as an instrument used by the analyst — not as the analyst itself.

Core Rule

EVA never trusts a model more than it trusts the evidence already observed in the dataset. If a model contradicts strong prior evidence, it is flagged for review — not deployed.

2.1 What This Means in Practice

- A slightly less accurate but explainable model may be preferred over a marginally better opaque one.
- Modeling is rejected when it would not influence a real decision.
- The system prefers no model over a misleading model.
- Machine learning is never activated for pure explanation, simple segmentation, descriptive reporting, or trend visualization.

3. When Machine Learning Is Activated

3.1 Activation Conditions

The MLRL activates only after all prior EVA layers have completed: Dataset Profiler & Semantic Understanding, Question Builder / Intent Inference, Data Repair & Integrity Layer, Feature Intelligence Engine, and Investigation & Hypothesis Engine.

Machine learning is triggered when the User Intent Record specifies any of the following:

- Prediction of a future outcome
- Forecasting over time

- Automated classification
- Risk scoring
- Detection of unseen or anomalous cases

3.2 Exclusion Conditions

Machine learning is explicitly not activated for:

- Pure explanation tasks
- Descriptive segmentation with no decision objective
- Descriptive reporting
- Trend visualization

⚠ Clustering Clarification

Clustering is a valid ML problem type (as a fallback when no explicit target exists but decision value remains). It is not the same as 'simple segmentation,' which refers to rule-based grouping for reporting purposes only. This distinction must be preserved throughout all module documents.

4. Global Analysis Ledger (GAL)

The Global Analysis Ledger is the single source of truth for the entire MLRL pipeline. No module communicates directly with another. All state, decisions, and reasoning flow through the GAL.

4.1 Purpose

- Makes every decision traceable and reproducible from one location
- Prevents modules from making assumptions not grounded in prior analysis
- Enables full audit of any modeling decision
- Allows individual modules to be re-run without breaking pipeline state

4.2 GAL Top-Level Structure

The following keys define the GAL's structure. Each module reads from and writes to specific keys only.

GAL Key	Contents
dataset_identity	Row meaning, domain, time behavior classification
user_intent_record	Objective, decision type, stakeholder type, interpretability priority
restricted_columns	Columns excluded from modeling (leakage, identifiers, post-outcome)

hypotheses	Key findings from the Investigation & Hypothesis Engine
feature_registry	Validated engineered features with reasoning
model_definition_record	Written by PFTDC — problem type, target, learning view schema, split strategy
candidate_model_record	Written by MCG — candidate approaches, metric, reasoning
model_evaluation_record	Written by TEM — performance, stability, generalization, selected model
model_validation_record	Written by RVEM — feature influence, hypothesis alignment, leakage check, validation status
prediction_deployment_record	Written by PMDD — deployment status, monitoring results, retraining flags
session_log	Timestamped record of all decisions, rejections, and reasoning across all modules

4.3 Read/Write Permissions by Module

Module	GAL Permissions
PFTDC	Reads: dataset_identity, user_intent_record, restricted_columns, hypotheses, feature_registry. Writes: model_definition_record
MCG	Reads: model_definition_record, user_intent_record, hypotheses. Writes: candidate_model_record
TEM	Reads: model_definition_record, candidate_model_record. Writes: model_evaluation_record
RVEM	Reads: model_evaluation_record, hypotheses, feature_registry. Writes: model_validation_record
PMDD	Reads: model_validation_record. Writes: prediction_deployment_record

5. Pipeline Architecture

5.1 Module Sequence

The MLRL consists of five coordinated modules that execute in sequence. Each module performs a reasoning task, not purely a computational one.

Module	Responsibility
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1. Problem Framing & Training Data Construction (PFTDC)	Determines if ML is appropriate. Defines the prediction task. Constructs the Learning View.
2. Model Candidate Generator (MCG)	Selects 2–4 appropriate modeling approaches based on problem type, data characteristics, and interpretability needs.
3. Training & Evaluation Module (TEM)	Trains each candidate. Evaluates performance, stability, and generalization. Selects the most reliable model.
4. Reliability & Validation (Explainability) Module (RVEM)	Verifies model reasoning aligns with prior evidence and hypotheses. Detects leakage. Assigns validation status.
5. Prediction & Monitoring (PMDD)	Deploys validated models as containerized services. Logs predictions. Monitors for drift and degradation.

5.2 Data Flow

GAL → PFTDC (Learning View) → MCG → TEM → RVEM → PMDD → Dashboard & Report

The pipeline operates on a Learning View — not the raw dataset. The Learning View is constructed from the repaired dataset, engineered features, and permitted columns only.

6. Pipeline Failure & Recovery

If any module rejects its output, the pipeline halts at that stage. The following recovery behaviors apply:

Failure Point	Recovery Behavior
PFTDC rejects modeling	Pipeline ends. Reason logged to GAL. EVA falls back to descriptive analysis. User is notified with explanation.
TEM rejects all candidates	Pipeline halts. GAL logs rejection reasons. PFTDC is notified to re-evaluate problem framing or adjust feature set before retry.
RVEM rejects model	Model is not deployed. RVEM logs contradiction. MCG and TEM may be re-run with revised constraints. If leakage is detected, PFTDC must re-run.
PMDD detects severe drift	Prediction service is flagged unreliable. Dashboard displays warning. New analysis session is requested. Automatic retraining does not occur.

⚠ Minimum Candidate Rule

If fewer than two candidates survive the MCG or TEM evaluation stages, the system must log a warning explaining why and proceed with a single candidate only if it passes all reliability checks. It must not silently reduce scope.

7. Class Imbalance Handling

Class imbalance is one of the most common real-world ML problems and must be handled explicitly. When the PFTDC detects severe class imbalance in the target variable, it must:

- Flag the imbalance in the `model_definition_record`
- Alert the MCG to adjust metric selection (e.g., prefer precision/recall over accuracy)
- Ensure the split strategy preserves class distribution across training, validation, and holdout sets

The MCG and TEM must acknowledge the imbalance flag and reflect it in candidate selection and evaluation criteria respectively.

8. Human Review & Override Touchpoints

EVA is a decision-support system. Human analysts retain authority over modeling decisions at the following points:

Touchpoint	Behavior
Conditional Validation	When RVEM assigns Conditionally Validated status, a human reviewer must acknowledge limitations before the model is used in high-stakes decisions. Dashboard displays a persistent caution indicator.
Contradiction Flag	When a model contradicts a strong prior hypothesis, it is flagged for human review. The system explains the contradiction but does not resolve it automatically.
Retraining Recommendation	PMDD recommends retraining but does not trigger it. A human must initiate a new analysis session.
Model Override	An analyst may override a model's prediction for a specific record. The override is logged in the <code>session_log</code> with reason.

9. Monitoring & Retraining Thresholds

The PMDD monitors deployed models continuously. The following thresholds define when alerts and retraining recommendations are triggered:

Drift Type	Threshold for Alert / Retraining Recommendation
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Confidence Drift	Average prediction confidence drops more than 15% relative to baseline over a rolling 100-prediction window.
Data Drift	Input feature distribution diverges from training distribution by a threshold equivalent to a population stability index (PSI) > 0.2 for any key feature.
Behavior Drift	The distribution of predicted outcomes shifts more than 20% relative to the baseline prediction distribution.
Error Pattern	If ground truth outcomes are available, model error rate exceeds 2x the holdout test error rate over any 200-prediction window.

These thresholds are defaults and may be adjusted in the session configuration by the analyst before deployment.

10. Conditional Validation — Operational Behavior

The RVEM assigns one of three validation statuses. The Conditionally Validated status requires specific operational behavior that must be enforced downstream:

Component	Behavior for Conditionally Validated Models
Dashboard	Displays a persistent orange caution banner on all predictions from this model. High-confidence predictions are still surfaced but marked.
Report Generator	Must include a dedicated section explaining what conditions are unresolved and what risks they pose.
Monitoring	PMDD runs monitoring checks at 2x the standard frequency for conditionally validated models.
Human Acknowledgement	For high-stakes use cases (as defined in <code>user_intent_record</code>), a human reviewer must log acknowledgement before predictions are acted upon.

11. Global Operational Rules

These rules apply across all modules and cannot be overridden at the module level:

1. ML cannot run without a defined target variable.
2. Restricted columns cannot be used at any stage of the pipeline.
3. All training steps, decisions, and rejections must be recorded in the GAL.
4. Models must pass Reliability & Validation before deployment.
5. Every prediction must include a confidence estimate.
6. No model may be deployed with a Rejected validation status.

7. The Learning View — not the raw dataset — is the only permitted training input.
8. Time-ordered datasets must use chronological splits. Random mixing is forbidden.
9. A simpler, reliable model is preferred over a complex, marginally better one.
10. If the pipeline fails, EVA must notify the user with a clear explanation.

12. Dockerized Service Architecture

Each MLRL module operates as an independent containerized service. This ensures reproducibility, failure isolation, independent re-execution, and scalability.

Services communicate through structured artifacts written to and read from the GAL. No module shares memory directly with another.

12.1 Session Artifact Structure

```
eva-session-data/sessions/<session_id>/dataset_snapshot/ | repaired_dataset/ |  
feature_view/ | learning_view/ | model_candidates/ | evaluation_results/ | selected_model/ |  
prediction_logs/ | ledger_snapshot/
```

All artifacts must be preserved to ensure full reproducibility. A model cannot be separated from its training metadata, feature schema, evaluation metrics, and reliability report.

13. Module Document Index

Each module is specified in a dedicated document. The following table maps modules to their documents and the GAL keys they own:

Module	Document
PFTDC — Problem Framing & Training Data Construction	EVA-ML-PFTDC.docx
MCG — Model Candidate Generator	EVA-ML-MCG.docx
TEM — Training & Evaluation Module	EVA-ML-TEM.docx
RVEM — Reliability & Validation (Explainability) Module	EVA-ML-RVEM.docx
PMDD — Prediction, Monitoring & Dockerized Deployment	EVA-ML-PMDD.docx

All module documents reference this master. In case of conflict between a module document and this master, this master takes precedence.